



1 RAPPORTAGE: NULINSPECTIE & EMC GAP-ANALYSE

Project: Simonshaven 380kV Uitbreiding (DP2-4)

Client: SC&M II V.O.F.

DNV Reference: OPP-00436810-PSP 25-1757

Date: 15-1-2026

Status: First Version

2 MANAGEMENT SUMMARY

This report documents the findings of the EMC Baseline Survey (*Nulinspectie*) performed by DNV at the Simonshaven 380kV substation.

Objective: To visually verify the "As-Built" EMC status of the existing Phase 1 (DP1) installation and identify specific integration risks for the new Phase 2-4 (DP2-4) expansion.

Key Conclusion:

Based on the review of the DP1 As-Built documentation [2], it is a known fact that the existing buildings (CDG and Takhuisjes) **do not** fully comply with the TenneT *Standard 380kV Zone Concept* [1] regarding building shielding (LPZ 2). This inspection serves to confirm the extent of these deviations.

Note: he existing buildings are confirmed to be unshielded (LPZ 0/1), the design for DP2-4 might be subject to higher immunity specifications or cabinet-level shielding for all new secondary equipment installed in these rooms. This will be evaluated during the development of the EMC management plan design.



3 INTRODUCTION

3.1 Project Context

SC&M is executing the expansion of the Simonshaven substation (DP2-4). This involves integrating new high-voltage assets (Transformers TR412-414, ~~Compensation Coils~~Shunt Reactors, Bays 4-5) into the existing operational environment (DP1).

3.2 Scope of Inspection

The scope of this *Nulinspectie* is limited to:

1. **Visual Inspection:** Verification of visible earthing, bonding, shielding, and cable routing in the existing CDG, Takhuisjes, and accessible cable trenches.
2. **Gap Analysis:** Comparing the physical installation against TenneT Standard E04 [1] and the DP1 EMC Plan [2].
3. **Interface Check:** Identifying physical connection points for the new DP2-4 earthing and cabling.

3.3 Dependencies & Limitations

Hidden Defects: DNV assumes that:

- U-underground earthing connections (not visible) were installed according to the as-built drawings, unless visual evidence suggests otherwise.
- The EMC mesh (not visible) was installed in the walls of the CDG according to the as-built drawings, unless visual evidence suggests otherwise.

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4 REFERENCE FRAMEWORK

The inspection evaluates the site against the following hierarchy of documents:

Ref	Document Number	Title	Status	Relevance
[1]	1287698 (E04)	<i>Uitwerking EMC eisen-zone concept standaard 380kV station</i>	The Standard	TenneT's mandatory rules for zoning (LPZ) and installation details.
[2]	2407-5582	<i>EMC beheersplan SMH380 (Phase 1)</i>	As-Built	Describes the executed state of DP1, including known deviations.
[3]	01350-01-00075	<i>Beoordeling EMC studie SMH380 Fase 1</i>	History	Previous gap analysis highlighting missing shielding mesh.

4.1 Risk Identification (Pre-Inspection Analysis)

Based on the review of Document [2] (*EMC beheersplan SMH380*), the following specific risks was important to be verified on-site:

1. **Legacy Shielding Defect:** Document [2] states that the EMC mesh (30x30mm) in the CDG and Takhuisjes was "not correctly installed."
 - Risk: The rooms may effectively be **LPZ 0** (Unshielded) instead of **LPZ 1**.
 - Inspection Focus: Verify if any wall/floor bonding points for the mesh are visible or connected.

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2. **Cable Entry Integrity:** There is ambiguity regarding the bonding of Roxtec frames and the termination of cable shields at the building entry.
 - *Risk:* High-frequency interference entering the building via cable shields.
 - *Inspection Focus:* Check physical bonding of frames to the earth bar (Detail 27 of [1]).
3. **PEC Continuity:** It is unconfirmed if Parallel Earthing Conductors (PECs) were installed in all concrete trenches.
 - *Risk:* Lack of shielding for field cabling.
 - *Inspection Focus:* Spot-check open trenches for 50mm² Cu PECs.

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5 INSPECTION FINDINGS

This section details the observations made during the visual inspection of the Simonshaven 380kV substation (DP1 area). The inspection focused on verifying the "As-Built" status against the TenneT *EMC eisen-zone concept* (Document E04) and identifying deviations reported in the Phase 1 documentation.

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5.1 Central Service Building (CDG)

5.1.1 Cable Entry Systems (Kabelinvoer)

- **Inspection Scope:** Visual check of the Roxtec cable transit frames in the AC/DC and Telecom rooms. Verification of frame bonding to the main earth bar and correct termination of cable shields (360° bonding) within the modules, in accordance with *TenneT Detail 27*.
- **Observations:**
 - The physical condition of the Roxtec frames was found to be **Poor**.
 - The bonding connection between the transit frame and the room's ring earth (aardrail) was **Missing**.
 - Regarding cable shields: **It was not possible to verify shield termination visually**.
- **Photo Reference:** SMH001-SHM016

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Photo reference SMH002

5.1.2 Room Shielding & Equipotential Bonding

- **Inspection Scope:** Verification of the Faraday cage concept (LPZ 2). Inspection of bonding points between the wall reinforcement (wapening/EMC-net) and the earthing system, and bonding of metal door frames.
- **Observations:**
 - Visible connections to the building reinforcement or EMC mesh were **Observed at regular intervals**
 - Metal door frames **Are** bonded to the structure earth with short flexible straps (liitze).

- **Photo Reference:** SMH019, SMH020



Photo reference **SMH019**

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5.1.3 Cabinet Installation (Secundaire Kasten)

- **Inspection Scope:** Check of cabinet earthing and cable gland plates (wartelplaten) in the CDG server/telecom room.
- **Observations:**
 - Cabinets are earthed via Short flat copper litz wire and Long round yellow-green wire (Pigtail) – this is found to be inconsistent and not a good practice. For new modifications, conform to TenneT standards more strictly.
 - Cable entry into cabinets is via Conductive gland plates with EMC glands : It was not possible to visually confirm the consistent implementation of ~~conductive-EMC~~ glands. For new modifications, conform to TenneT standards ~~more strictly~~.
- **Photo Reference:** SMH011, SMH012



Photo reference SMH012

5.2 Bay Houses (Takhuisjes)

5.2.1 Building Shielding Integrity Takhuisje 2

- **Inspection Scope :** Verification of the "Missing Mesh" deviation noted in previous reports. Check for visible earth bars connected to the wall structure.
- **Observations:**

- The inspection **Could not confirm** the presence of the 30x30mm EMC mesh described in the design.
For new modifications, conform to TenneT standards ~~more strictly~~.
- Earth bars (aardrails) on the walls **Are** connected to the building reinforcement at the required intervals (max 5m). The detailed measurements between the connections were not taken but **For new modifications, conform to TenneT standards ~~more strictly~~.**

5.2.2 Secondary Equipment Earthing Takhuisje 2

- **Inspection Scope:** Check of protection relay cabinets and auxiliary power distribution boards.
- **Observations:**
 - Earthing of the cabinets to the room ring earth is **Inconsistent**. **For new modifications, conform to TenneT standards ~~more strictly~~.**
- **Photo Reference:** SMH036

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Photo reference SMH036

5.3 Field Infrastructure (Buitenterrein)

5.3.1 Concrete Cable Trenches (Kabelgoten)

- **Inspection Scope:** Covers were not lifted but presence of a Parallel Earthing Conductor (PEC) was visible and terminated inside the Takhuisjes.
- **Observations:**
 - **Location 1 (Near CDG):** A copper PEC ($\geq 50\text{mm}^2$) **Was** visible terminating inside. No External PEC was identified.
 - **Location 2 (Near Tak 1):** A copper PEC **Was** visible inside all Tak-huisjes
- **Photo Reference:** IMG_6485



Photo reference IMG_6485

5.3.2 Primary Equipment Bonding

- **Inspection Scope:** Not all structures were inspected as this is an earthing and bonding scope of work and not EMC. However, Visual check of earthing connections of the lightning termination rod were captured (TenneT requirement: 2x connections for main structures).
- **Observations:**
 - The condition of the connections is **Good**
- **Photo Reference:** SMH048



Photo reference SMH048

5.3.3 Fence Earthing (Hekwerk)

- **Inspection Scope:** Verification of earthing straps between fence panels and connection to the station earth grid to prevent touch voltage hazards.
- **Observations:**
 - Flexible earthing straps between fence panels are **Missing in some sections / Broken**.
 - Connection to the main earth grid was **Not visible in all locations**.
- **Photo Reference:** SMH052-SMH061



Photo reference SMH054

5.4 General Observations

Housekeeping: The general level of EMC housekeeping (e.g., loose wire clippings, metal debris in cabinets) **Needs Improvement.**

6 GAP ANALYSIS & RECOMMENDATIONS FOR DP2-4

6.1 Confirmed Deviations

Based on the inspection, the following deviations from TenneT Standard [1] are confirmed in the existing DP1 installation:

- **Deviation 1:** CDG Building Shielding is ineffective until proven through measurements that effective shielding is indeed present.
- **Deviation 2:** Takhuisjes have various findings. Secondary cabinets are not properly connected with metallic glands with the exception of Takhuisje 3. Excessive cabling are-is already installed, especially at Takhuisje 1 – It will be very difficult practically to install new cables. New Takhuisje EMC design to conform to TenneT standards more strictly.

6.2 Implications for DP2-4 Design

To ensure EMC compliance for the new expansion despite these legacy issues, DNV recommends the following strategy for the DP2-4 design:

1. **Zoning Strategy:** The EMC Management Plan for DP2-4 will classify the existing buildings as **LPZ 0**.
2. **Mitigation:** New secondary equipment for DP2-4 installed in these buildings must rely on **Cabinet-Level Shielding** (Metal housings, EMC glands) rather than Room-Level shielding.
3. **Retrofit (If required):** Investigate the challenge of installing new cables in Tak 1 and if any retrofits are required to ensure EMC and practical installation requirements.

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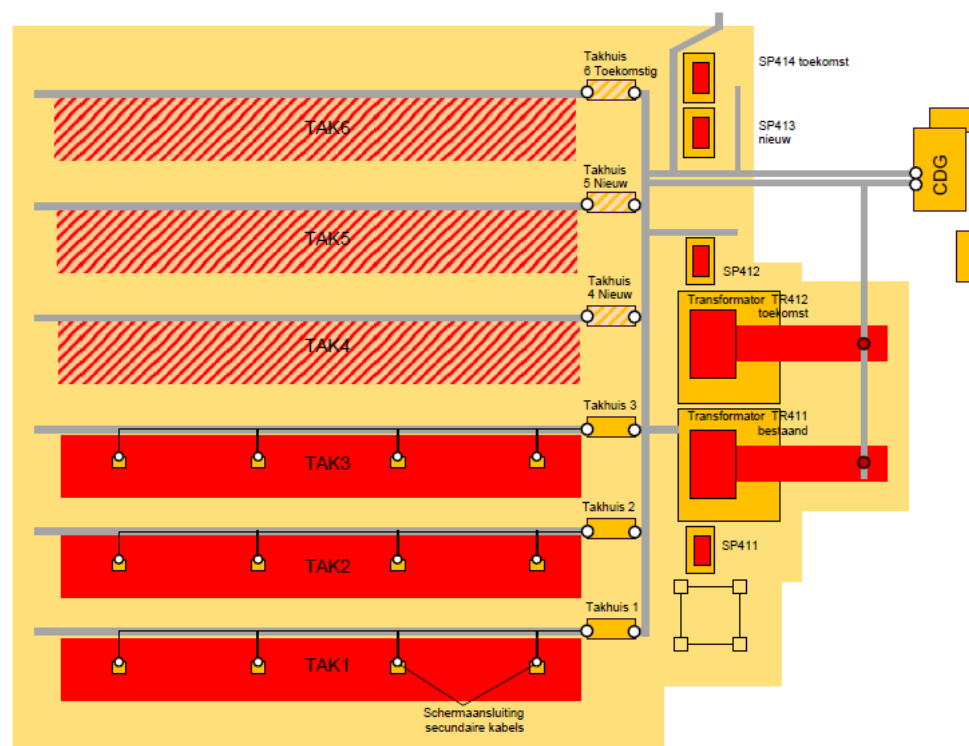


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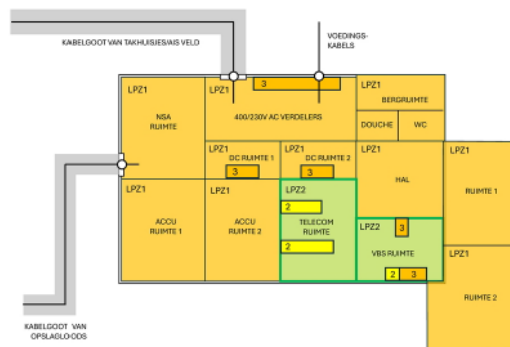
Bijlage A EM zone indeling SMH380

LEGENDA

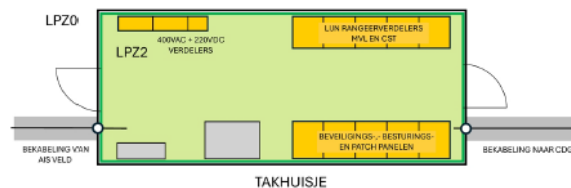
- 2** = Apparatuur in takhuisjes en het CDG die beschermd wordt door extra afscherming, apparatuur in telecomruimte die geen industriële immuniteitspecificaties heeft.
- 3** = industrieel, secundaire installatie in takhuisjes en CDG en langs hekwerken rond de AIS schakeltuin.
- 4** = locaties binnen de primaire schakeltuin, ter plaatse van railsystemen, schakelaars, scheiders, transformatoren, spoelen en kabeleindsluitingen.
- = EM/LPZ zone overgang d.m.v. kabelschemaansluiting secundaire kabels bij invoer veldkasten, takhuisjes en CDG
- = Eindsluitingen HS kabels



Figuur A.1 EM zone indeling SMH380



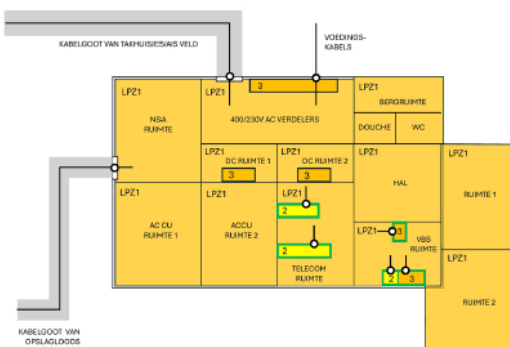
EM zoning CDG conform TenneT eisen



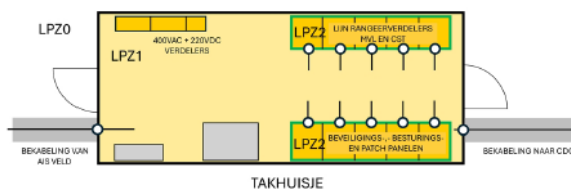
EM zoning takhuisje conform TenneT eisen

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-  = LPZ1, afscherming door middel van betonwapening overeenkomend met EM3
-  = LPZ2, afscherming door middel van EMC-net 30x30x3mm
-  = EM2 apparatuur
-  = EMC kabeldoorvoer met scherm aansluiting



EM zoning CDG conform zoals uitgevoerd



EM zoning takhuisje conform zoals uitgevoerd

Figuur A.2 EM en LPZ zone indeling CDG en takhuisjes